

35KW Body Maximum Power Setup and Calibration

([Click here for Summary Version](#))

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	45 mins	Not Applicable

2 Overview

Reference Material:

- [Keyboard Operation / Description of IPA and PA Stages](#)
- [WA Setup and ATTN Setup Keypad Controls](#)

Last Update	03/02/2005
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3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Power Measurement Kit (Wattmeter Kit)	1	-	2372868	-
Medium Phillips Screwdriver	1	-	-	-
12 in Crescent Wrench	1	-	-	-

3.2 Safety



DANGER

FATAL ELECT/RIC SHOCK HAZARD

THE RF AMPLIFIER IS NOT DISABLED

TO PREVENT FATAL ELECT/RIC SHOCK, ENSURE THE RF AMPLIFIER IS DISABLED WHILE CONFIGURING THE CALIBRATION TOOLS.



NOTICE

Property Damage! Prevent Coil And Associated Switch Damage, By Removing All Phantoms And Hardware (I.E., Head Coil, Surface Coil...) From The Magnet Bore.



NOTICE

Possible equipment damage. Do not operate the RF amplifier with an improper load connected to its output. Doing so may permanently damage the amplifier.



NOTICE

Possible equipment damage. Do not turn connect/disconnect Wattmeter Sensor while Meter is ON. Always turn Meter OFF prior to plugging/unplugging Sensor.

3.3 Required Conditions

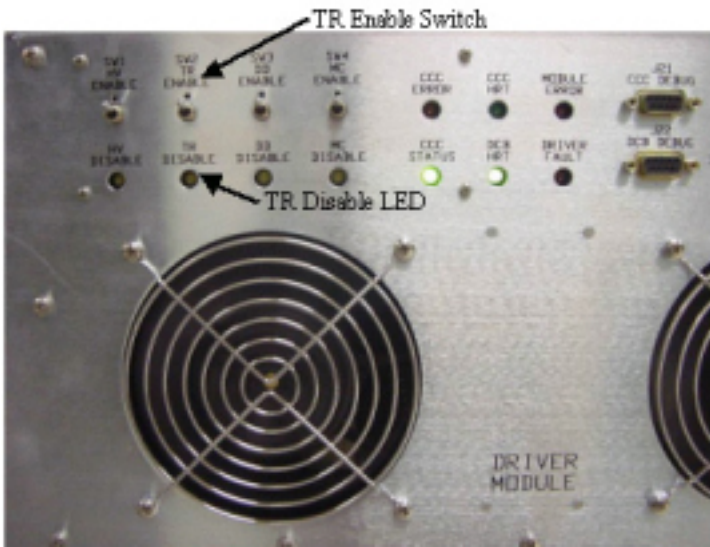
Condition	Reference	Effectivity
Verify that the system is idle and all coils have been removed from magnet bore.	-	-

4 Procedure

4.1 Production Systems Using TR In Driver Module

1. Verify that the system is idle and all coils have been removed from magnet bore.
2. Remove the front cover of the EXCITE Systems Cabinet.
3. On the front of the Driver Module, disable T/R error reporting by setting the TR switch to the DIS (disable faults) position. (see [Illustration 1](#)).

Illustration 1: Driver Module Switch

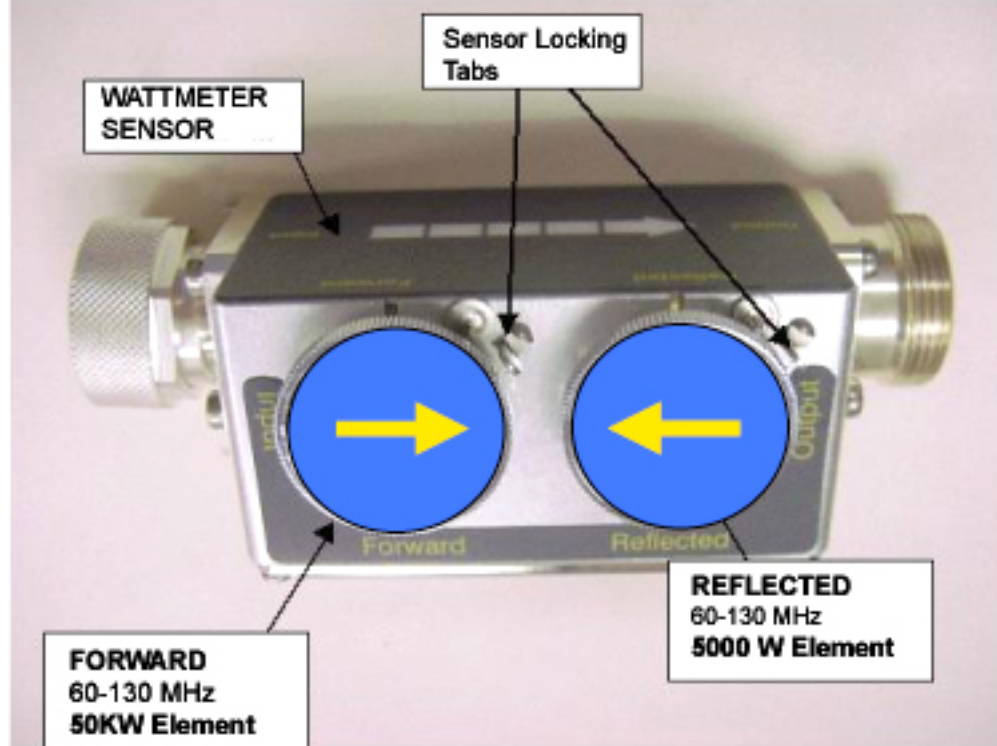


NOTE: HV, DD and MC should remain set to Enable.

4.2 Setting up the RF Coupler

1. Insert the 50KW Element into the Forward Sensor Port, (Blue- 4300A374-3). Make sure the arrow on the element points in the direction of the arrow (from Input to Output) on the Sensor. Lock Element in place using the locking tab on the sensor. ((see [Illustration 2](#)))

Illustration 2: Rf Coupler Setup For Body Mode



2. Insert the 5000W Element into the Reflected Sensor Port (Blue- 4300A374-6). Make sure the arrow on the element points in the opposite direction of the arrow (from Input to Output) on the Sensor. Lock Element in place using the locking tab on the sensor. ((see [Illustration 2](#))) The reflected port is not used during this procedure. Placing the 5000W element into the reflected port is just a safety measure, and is not used during Body calibration.

4.3 Wattmeter and Dummy Load Setup



WARNING

ELECTRICAL/RF BURN HAZARD

RF BURNS AS A RESULT OF DISCONNECTING HELIAX CABLES WHILE THE SYSTEM IS SCANNING PREVENT POSSIBLE RF BURNS WHEN DISCONNECTING HELIAX CABLES FROM J3 OR J4 ON THE RF AMP BY VERIFYING THAT THE SYSTEM IS NOT MANUALLY PRESCANNING OR SCANNING. VERIFY THAT THE SCAN DESKTOP ICON DISPLAYS THE "IDLE" MESSAGE AND THE RF ENABLE SWITCH IS IN THE "OFF" POSITION.

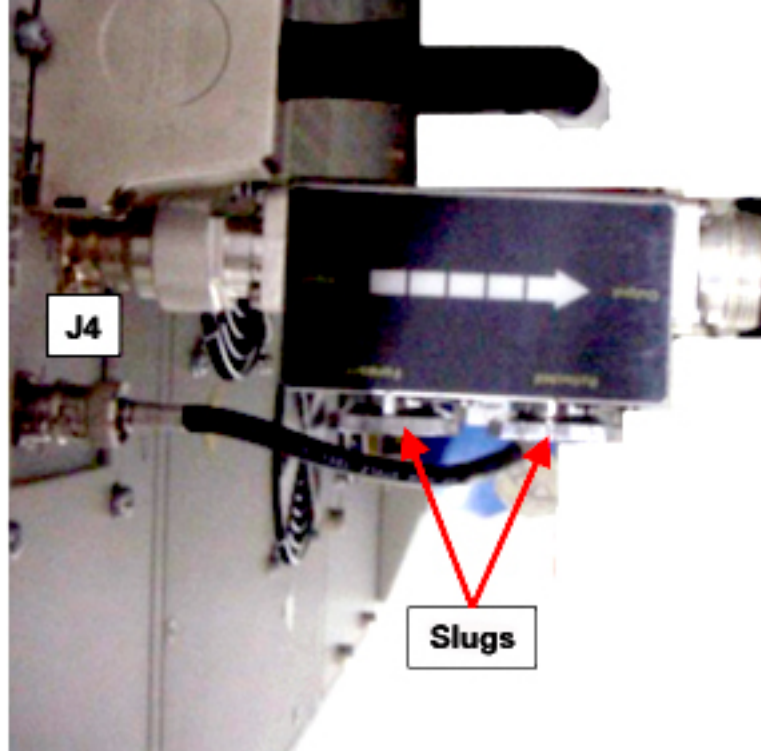


NOTICE

Possible equipment damage. Do not turn connect/disconnect Wattmeter Sensor while Meter is ON. Always turn Meter OFF prior to plugging/unplugging Sensor.

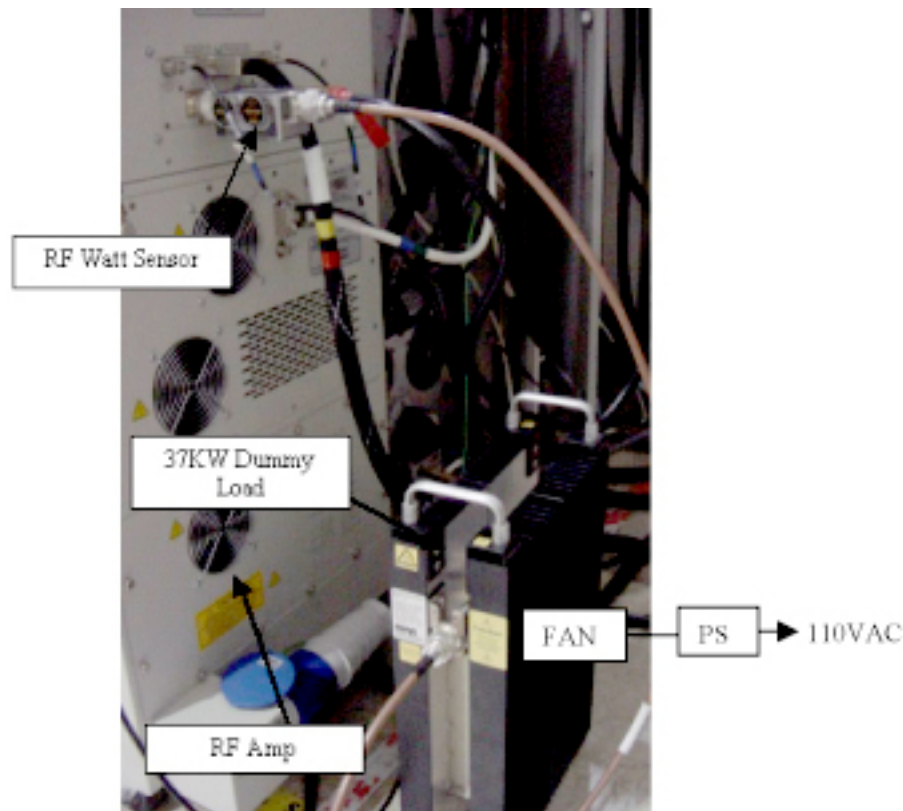
1. Verify scanner is in "idle", turn off RF Enable at the Exciter (FP-I/O) or disconnect J14 RF Drive Input) from the back of the RF amplifier.
2. Disconnect the Body cable J4 "Body Output" from the back of the RF Amplifier.
3. To protect the Head Amplifier section, disconnect the Head cable J3 "Head Output" from the back of the RF Amplifier as well.
4. Connect the Input of the Coupler to J4 on the back of the RF Amplifier. ((see [Illustration 3](#))).

Illustration 3: Connection Of RF Coupler To Body Output



5. Connect the Output of the RF Coupler to the 35KW Dummy Load ([Illustration 4](#)).

Illustration 4: Complete Body Output Test Setup



6. Connect one end of the Wattmeter cable to the Wattmeter Sensor connector, (RS232 is unused). Connect the other end to the Coupler.
7. Make sure to plug in the fan for the dummy load. AT NO TIME should you use this dummy load without its fan running.

NOTICE

Do Not Apply RF Power To The Dummy Load Without Its Fan In Operation. Doing So Will Result In Catestrphic Damage To The Dummy Load.

8. Reconnect J14 on the back of the amplifier. Switch RF Enable to **ON**.

4.4 Setting Up The Wattmeter to Measure Forward Body Maximum Power

1. For a more detailed discussion of the Wattmeter, see [Watt Meter \(2372868\) Introduction](#).
2. Press the **ON** button on the Wattmeter. (If coupler and sensor is not correctly attached to the amplifier you will get a “No Sensor” indication)
3. Select the arrow $\triangle$ button under Scale.
4. Type: **50000** to set the range.
5. Select <Enter> button.
6. Check the setting, by selecting the arrow $\triangle$ button under Scale once again.
7. Press <Enter> to exit this mode.
8. Select the arrow $\triangle$ button under *Fwd Units* make sure the scale reads “w” (or “kW “on some watt meters). If the scale reads dBm, select the arrow $\triangle$ button under *Fwd Units* again, to toggle the scale to read “w” (Watts or kW kiloWatts) on the TOP, Forward Power scale. Ignore the BOTTOM, reflected Power scale value. It is not used during Maximum Power Calibration.
9. Select the <Shift> button.
10. Select the arrow $\triangle$ button under Setup.
11. Verify it reads “43”. If not, type the number “43” followed by the <Enter> button. ([Illustration 5](#) shows what should be displayed at this time).

Illustration 5: Wattmeter Display Set To Read Peak Power Sensor



NOTE: This action displays the Sensor Type configured in the Wattmeter. This procedure uses Sensor” Type 43”, which is optimized for peak power measurement.

12. Select <Enter> button.
13. Select the arrow $\triangle$ button under *Type*, change unit of measurement to Peak.
14. Some kits require an offset to ensure accurate measurement. Locate the Element Offset Factors card in the kit and view the offset for 3T Body Output. See [Illustration 6](#).

Illustration 6: Element Offset Factor Card

Power Measurement Quick Reference Card for GE3T Kit
Element Offset Factors

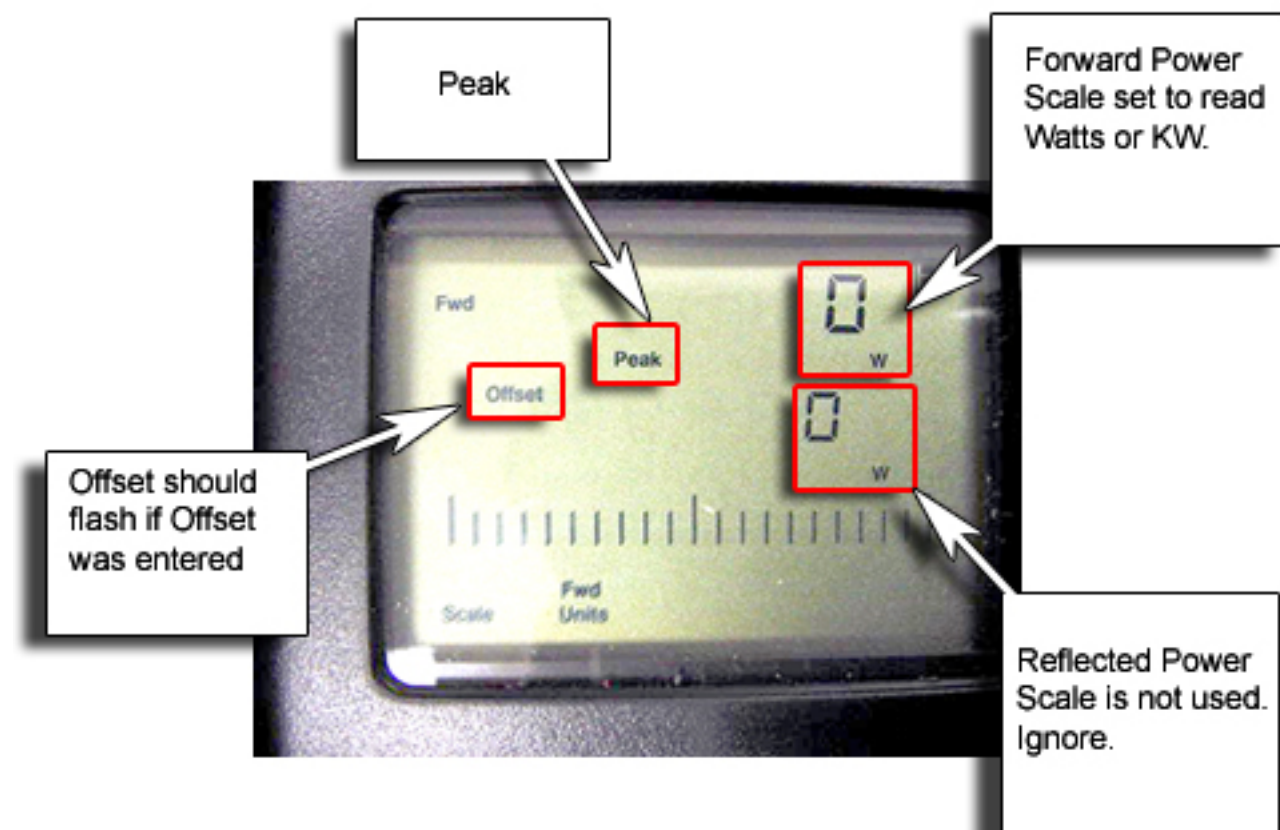
Description	Frequency (MHz)	Power Level (Watts)	Element	Offset
3 T Body Output	127.72	35000	4300A374-3	-0.1128
3 T Head Output	127.72	4000	4300A374-6	-0.0216
0.35 T Body Output	14.85	5850	4300A374-1	0.1506
0.35 T Head Output	14.85	620	4300A374-4	-0.0435
1.5 T Body Output	63.86	16000	4300A374-2	-0.0431
1.5 T Head Output	63.86	2000	4300A374-5	-0.1427
1.0 T Head Output	42.00	750	4300A374-5	-0.3028
1.0 T Body Output	42.00	7500	4300A374-2	-0.3438
3 T MVS 4k Output	51.70	4000	4300A374-5	-0.0298
3 T MVS 8k Output	51.70	8000	4300A374-2	-0.5538
0.7 T Head Output	29.80	950	4300A374-5	-0.3486
0.7 T Body Output	29.80	8000 2000	4300A374-2	-0.4349

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Offset for 3T Body Output (EXAMPLE card only)

15. If an offset exists, press the Offset Button on the wattmeter. Enter in the value that was displayed on the kits offset card and press enter. (Illustration 7 shows what should be displayed at this time).

Illustration 7: Wattmeter Set To Display Peak Power



4.5 Scan Setup

1. Prepare the system to scan in Body mode:
 - a. Click on [New Pt]
 - b. Id: **geservice**<Enter>
 - c. Name: **apb cal body**<Enter>
 - d. Weight (Lb): **300**<Enter>
 - e. Set Patient Protocols to [Service]
 - f. Scroll through the protocol field until you find the protocol [0.41-3.0T apb cal].
 - g. Select [Series 1, Body]

h. [Accept]

2. Set **Landmark**

3. [Save Series]

4. Select [Research Options] with the mouse cursor, and right click. Then select [Setup Params]. Set *TG* to **50**[Done].

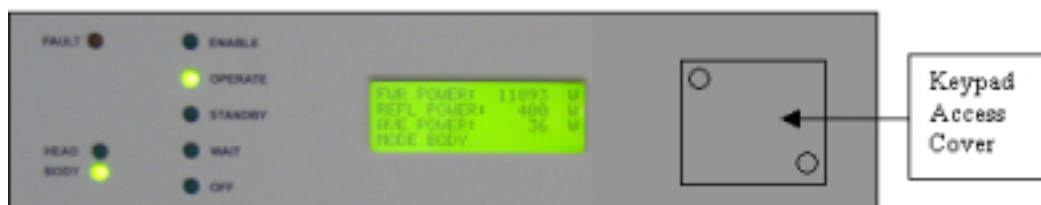
5. Select [Research Options] with the mouse cursor, and right click, select [Display CVs] From the drop down menu.

6. Set value of CV *calmode* to **5**<Enter> (Dual Logamp Waveform). (Make sure to press <Enter> to capture the setting.

7. Select [Accept], then [Download].

8. Go to the amplifier and remove the Keypad cover, located on the front of the amplifier near the display. (see [Illustration 8](#)).

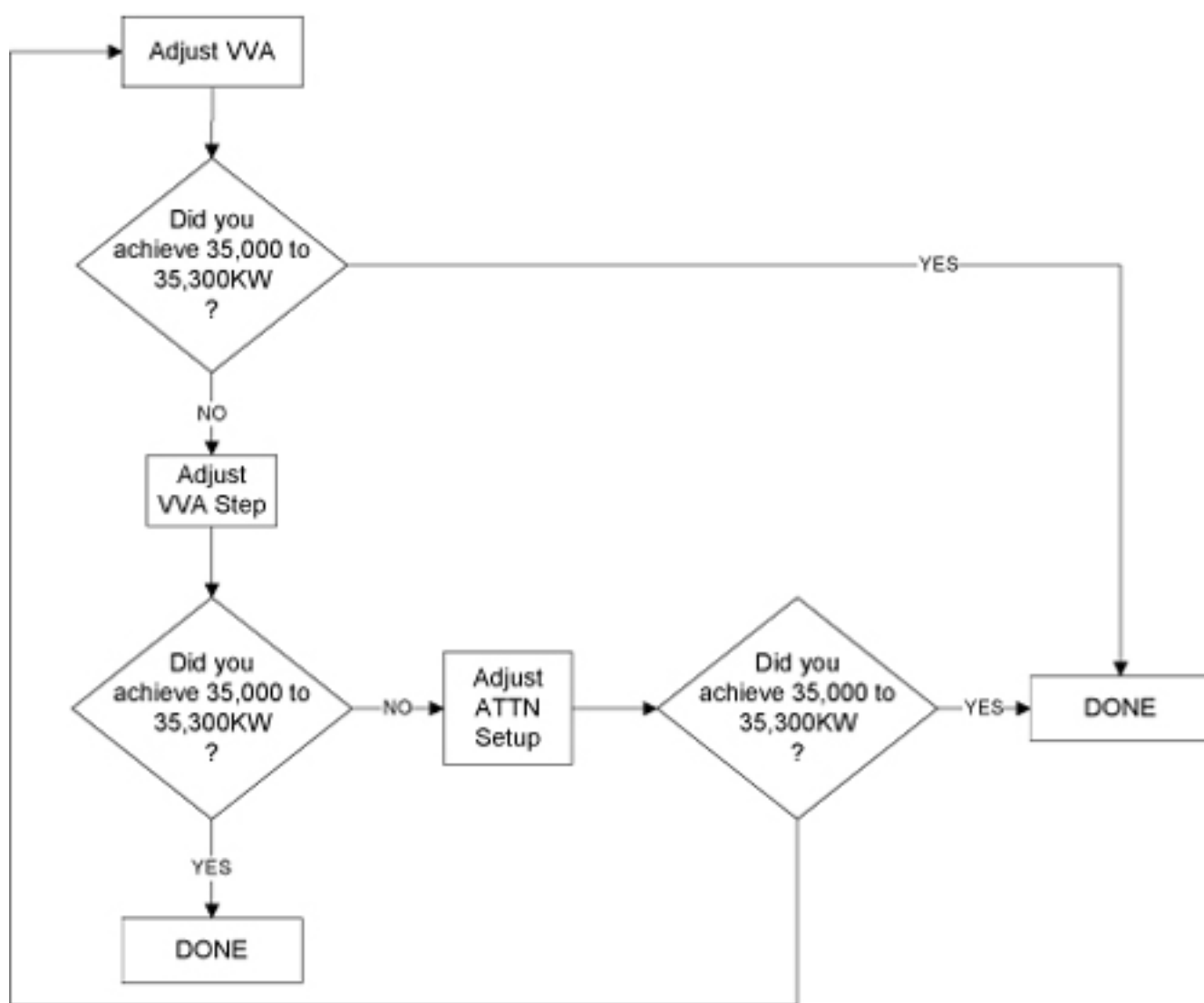
Illustration 8: FRONT OF RF AMP CONTROL MODULE



4.6 Adjusting the RF Amplifier Output Power (Body)

1. This section adjusts the gain of the Solid State pre-amplifier (VVA SETUP) and if necessary, also adjusts the Attenuation of the PA stage, (ATTN SETUP). The Gain (VVA -Variable Voltage Adjustment) of the Solid-state amplifier is setup first, maximizing its output. The Attenuation of the PA stage can also be adjusted to “Fine tune” the final value to as close as possible to 35KW, (34,500-35,300 Watts). VVA Setup and ATTN Setup interact with each other, and you may find yourself jumping back and forth between these adjustments until the amplifier is properly calibrated.

Illustration 9: Process Flow



2. Select [Manual Prescan].
3. Select [Scan TR].
4. Set the Transmit Gain to **180**. Make sure everything is functioning normally and you are getting a reading on the Wattmeter. Allow scanner to pulse at TG180 for 6 minutes. This is due to the drop in gain that occurs over the 6 minute period.
5. Slowly increase the TG to **200**. (If the amplifier trips set the TG at its highest setting).

4.7 VVA SETUP (Solid State amplifier Gain adjust)



NOTICE

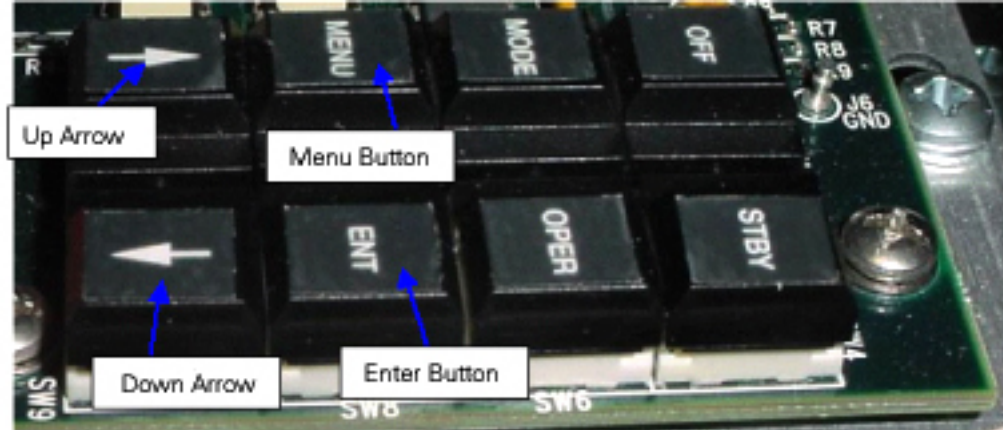
Press Only The Buttons Required By This Procedure. Avoid Pressing Mode Button. It Controls The Output Drive Of The Solid State Amplifier, Switching From Body To Head Mode Of Operation. See [Keyboard Operation / Description of IPA and PA Stages](#) for information on the RF Setup Keypad.

1. While the system is pulsing RF, Press the <Menu Button> on the Keypad located inside the RF amplifier, ([Illustration 10](#)) until the LCD Display on the front of the amplifier reads the following:

> ATTN SETUP

VVA SETUP

Illustration 10: RF Setup Keypad



2. Press <Down Arrow> to select VVA SETUP.

ATTN SETUP

>VVA SETUP

3. Press <Enter>

VVA STEP: 1 (Each time Enter is pressed this incremental Step Value changes)

VVA: 3368

4. Continue to press <Enter> button until you've. selected the desired step size. For smaller and finer calibration use smaller incremental step sizes such as 1 or 10. (Step size values can be set to 1,10,20,30,40,50,60,70,80,90,or 100).
5. Once the Step Size is selected, Press <Down> or <up> arrow to actually adjust the Gain:
VVA Step: 10
>VVA: 3368 (This value will change by 10 counts)
6. Using the <Down>/<Up> arrows resume the adjustment of the gain until you have achieved 35KW (Approx. 34,500-35,300 Watts on the meter).
7. If you can achieve 35KW (Approx. 34,500-35,300 Watts) on the meter display Press the <Menu> button until you reset the keypad to its power measurement display.

NOTE: If you are still not able to achieve the proper maximum Wattage setting, you may need to adjust the PA stage Attenuation, see [Adjusting ATTN SETUP \(PA Attenuation\)](#), and then repeat [VVA SETUP \(Solid State amplifier Gain adjust\)](#).

8. Proceed to [Saving New Calibration Settings](#) Saving New Calibration Settings.

4.8 Adjusting ATTN SETUP (PA Attenuation)

This section only needs to be performed if you can't achieve 35KW (Approx. 34,500-35,300 Watts) with VVA SETUP. (Note: At this time the RF amplifier is still pulsing at TG of 200)

1. Press the Menu Button until the LCD Display on the front of the amplifier reads the following:

>ATTN SETUP

VVA SETUP

2. Select <Enter>.

> ATTN: 13 (This value will change when the down/up arrows are pressed)

3. Press the <Down> or **Up** arrows while watching the Wattmeter to adjust the attenuation of the RF amplifier PA

stage. Slowly attenuate the RF power to as close to 35KW as you can. (Approx. 34,500-35,300 Watts). If you cannot achieve 35KW, adjust the attenuation to a value that is below 35KW.

4. Press the <Menu> button to exit
5. Press the <Menu> button again to return the LCD Display to its power measurement display.
6. If you did not achieve 35KW, proceed to [VVA SETUP \(Solid State amplifier Gain adjust\)](#) VVA SETUP (Solid State amplifier Gain adjust). If you achieved 35KW (Approx. 34,500-35,300 Watts) proceed to [Saving New Calibration Settings](#) Saving New Calibration Settings.

4.9 Saving New Calibration Settings

1. As a quick check, look at the LCD display on the front of the amplifier. It should be reading 35KW + or - approximately 2000 Watts. This will vary site to site and can be used as a confidence indicator as to your actual setting. The LCD display should not be relied on as an accurate measurement device.
2. Stop [Manual Prescan].
3. Press the <Menu> button multiple times until the original screen on the amp LCD reappears.
4. Press the **Off** button. This will STORE the value in NVRAM. Wait 30 seconds and press the "Operate" (Or "standby") button. Wait another 30 seconds for the amplifier to react to the new settings.
5. Reduce the TG to zero, once the RF Output is calibrated to specification. If the Amplifier is setup correctly, the RF amplifier Display, and Wattmeter should be very close to the same value.
6. Test the new Maximum Power value by restarting [Manual Prescan].
7. Select [Scan TR].
8. Slowly increase the TG to **200**. The system should not trip.
9. Take a final look at the Wattmeter display; making sure the desired value is displayed.
10. Reduce TG to Zero.
11. Stop [Manual Presecan]
12. Continue with Finalization [Step](#).

5 Finalization



WARNING

ELECTRICAL / RF BURN HAZARD

**RF BURNS AS A RESULT OF DISCONNECTING HELIAX CABLES WHILE THE SYSTEM IS SCANNING
PREVENT POSSIBLE RF BURNS WHEN DISCONNECTING HELIAX CABLES FROM J3 OR J4 ON THE
RF AMP BY VERIFYING THAT THE SYSTEM IS NOT MANUALLY PRESCANNING OR SCANNING.
VERIFY THAT THE SCAN DESKTOP ICON DISPLAYS THE "IDLE" MESSAGE AND THE RF ENABLE
SWITCH IS IN THE "OFF" POSITION.**



NOTICE

Property Damage! Prevent Coil And Associated Switch Damage, By Removing All Phantoms And Hardware (I.E., Head Coil, Surface Coil...) From The Magnet Bore.

1. Before proceeding, perform a [Reset TPS]. This is to ensure that the RF Amplifier is in a known state to the system.
2. Perform UPM ([*Universal Power Monitor - Body Setup and Calibration*](#)) to ensure the UPM is properly calibrated and functioning properly.
3. Verify scanner is in “idle”, turn off RF Enable and disconnect J14 from the back of the RF amplifier.
4. Disconnect the RF Watt Sensor from the J4 Body Output Connection.
5. Connect the Original Body Output Cable to RF Amplifier, J4 and original head output cable to RF amp J3.

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